

Comprehensive Analog Electronics Notes

<https://gemini.google.com/app/edd453cd1a690c03>

User prompt: Basic theory and operation of PNP and NPN transistors, Characteristics of Common Base, Common Emitter and Common Collector configuration, DC Biasing: Fixed Bias, Emitter Bias, voltage divider bias, Field effect transistor: JFET, Drain and Transfer characteristic, MOSFET, Introduction to Operational Amplifier and its Applications as Adder, Subtractor, Integrator, Differentiator, log antilog. Give proper note in extended form without missing anything Give it pdf to me

Response: Comprehensive Engineering Notes: Analog Electronics

1. Bipolar Junction Transistors (BJT): Theory and Operation

A Bipolar Junction Transistor (BJT) is a three-terminal, three-layer, two-junction semiconductor device. It is termed **bipolar** because conduction involves both charge carriers: electrons (negative carriers) and holes (positive carriers).

Physical Construction and Layer Characteristics

A BJT consists of three distinct regions:

1. Emitter (E):

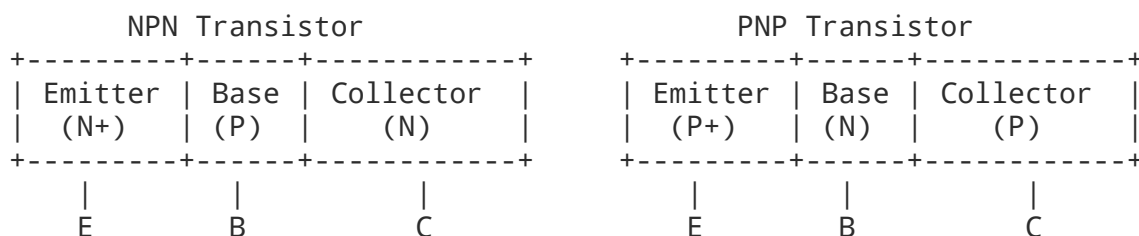
- **Doping:** Heavily doped.
- **Size:** Moderately sized.
- **Function:** Injects majority carriers into the base.

2. Base (B):

- **Doping:** Very lightly doped.
- **Size:** Extremely thin (typically on the order of a few micrometers).
- **Function:** Passes majority of injected carriers through to the collector with minimal recombination.

3. Collector (C):

- **Doping:** Moderately doped (lower than emitter, higher than base).
- **Size:** Physically largest region to facilitate heat dissipation.
- **Function:** Collects carriers arriving from the emitter via the base.



Modes of Operation

A BJT has two pn junctions: the **Emitter-Base Junction (EBJ)** and the **Collector-Base Junction (CBJ)**.

Operating Mode	EBJ Bias	CBJ Bias	Primary Application
Cutoff	Reverse	Reverse	Digital Switch (OFF state)
Active	Forward	Reverse	Linear Signal Amplification
Saturation	Forward	Forward	Digital Switch (ON state)
Inverted (Reverse Active)	Reverse	Forward	Specialized low-gain switching

Operation of NPN Transistor (Active Mode)

- Biasing Setup:** EBJ is forward-biased ($V_{BE} > 0$, typically $\approx 0.7 \text{ V}$ for Silicon), and CBJ is reverse-biased ($V_{CB} > 0$).
- Carrier Injection:**
 - The forward bias on the EBJ lowers the potential barrier, causing large numbers of majority electrons from the heavily doped n^+ -emitter to diffuse into the p -type base.
 - Simultaneously, a small number of holes from the base are injected into the emitter, constituting a minor component of the emitter current.
- Base Transit and Recombination:**
 - Because the base is very thin and lightly doped, only about 1% to 5% of the injected electrons recombine with holes in the base. This small recombination creates the external base current (I_B).
- Collection:**
 - The remaining 95% to 99% of electrons diffuse across the base to reach the edge of the depletion region of the reverse-biased CBJ.
 - The electric field across the reverse-biased CBJ accelerates these electrons into the collector region, forming the collector current (I_C).

Operation of PNP Transistor (Active Mode)

- Biasing Setup:** EBJ is forward-biased ($V_{EB} > 0$, typically $\approx 0.7 \text{ V}$), and CBJ is reverse-biased ($V_{BC} > 0$).
- Carrier Injection:** The p^+ -emitter injects majority holes into the thin, lightly doped n -type base.
- Base Transit and Recombination:** A small fraction of holes recombine with electrons in the base, creating a small external base current entering/leaving the terminal based on polarity.
- Collection:** Most holes cross the base into the CBJ depletion region, where the strong electric field sweeps them into the p -type collector region.

Fundamental Current Relationships

By Kirchhoff's Current Law (KCL):

$$I_E = I_B + I_C$$

Common Base Current Gain (α)

α represents the fraction of emitter current that reaches the collector:

$$\alpha = \frac{I_C}{I_E} \quad (\text{typically } 0.95 \leq \alpha \leq 0.998)$$

Taking into account the collector reverse saturation current (I_{CBO}):

$$I_C = \alpha I_E + I_{CBO}$$

Common Emitter Current Gain (β)

β (or h_{FE}) is the ratio of collector current to base current:

$$\beta = \frac{I_C}{I_B} \quad (\text{typically } 20 \leq \beta \leq 500)$$

Interrelationships Between α and β

$$\beta = \frac{\alpha}{1 - \alpha}$$

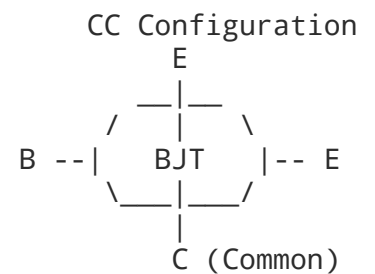
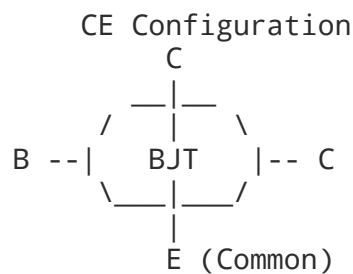
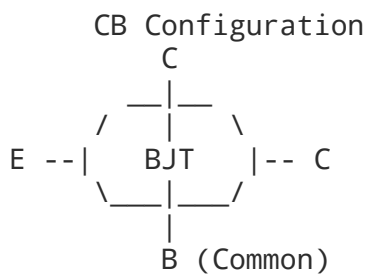
$$\alpha = \frac{\beta}{\beta + 1}$$

Taking into account leakage currents:

$$I_C = \beta I_B + I_{CEO} = \beta I_B + (\beta + 1)I_{CBO}$$

2. Transistor Configurations and Characteristics

Depending on which terminal is shared between input and output, the BJT can be connected in three primary configurations: **Common Base (CB)**, **Common Emitter (CE)**, and **Common Collector (CC)**.



Common Base (CB) Configuration

- **Input Terminal:** Emitter
- **Output Terminal:** Collector

- **Common Terminal:** Base (AC ground)

1. Input Characteristics (I_E vs. V_{EB} at constant V_{CB})

- Resembles a forward-biased pn junction diode.
- Once V_{EB} exceeds the cut-in voltage ($V_\gamma \approx 0.7$ V for Si), I_E increases exponentially.
- As reverse bias $|V_{CB}|$ increases, the depletion layer of the CBJ widens into the base (**Early Effect** or base-width modulation). The effective base width decreases, reducing recombination and slightly shifting the curves leftward for a given V_{EB} .

2. Output Characteristics (I_C vs. V_{CB} at constant I_E)

- **Active Region:** The collector-base junction is reverse-biased ($V_{CB} > 0$). Curves are nearly horizontal lines, showing that $I_C \approx \alpha I_E$, practically independent of V_{CB} . Output resistance is very high ($r_o \approx M\Omega$).
- **Saturation Region:** Occurs when $V_{CB} < 0$ (the CB junction becomes forward-biased). Carriers are repelled, and I_C drops sharply to zero.
- **Cutoff Region:** $I_E = 0$; only a minute leakage current $I_C = I_{CBO}$ flows.

Common Emitter (CE) Configuration

- **Input Terminal:** Base
- **Output Terminal:** Collector
- **Common Terminal:** Emitter

1. Input Characteristics (I_B vs. V_{BE} at constant V_{CE})

- Represents forward-biased EBJ diode characteristics.
- As V_{CE} increases, the collector depletion layer widens, reducing effective base width. Less recombination occurs, so fewer base carriers are needed to support collector current; hence, for a fixed V_{BE} , I_B decreases slightly (curves shift slightly to the right).

2. Output Characteristics (I_C vs. V_{CE} at constant I_B)

- **Active Region:** $V_{CE} > V_{CE(sat)} \approx 0.2$ V. $I_C = \beta I_B$. Curves exhibit a slight upward slope due to the **Early Effect** (base-width modulation). Extrapolating these lines backward intersects the negative voltage axis at the **Early Voltage** ($-V_A$).

$$r_o = \left. \frac{\Delta V_{CE}}{\Delta I_C} \right|_{I_B = \text{const}} = \frac{V_A + V_{CE}}{I_C} \approx \frac{V_A}{I_C}$$

- **Saturation Region:** $V_{CE} < V_{CE(sat)}$ (0.1 V to 0.3 V). Both EBJ and CBJ are forward-biased. I_C drops sharply and is limited strictly by external load elements ($I_{C(sat)} < \beta I_B$).

- **Cutoff Region:** $I_B = 0$. EBJ is no longer forward-biased. A small residual collector-to-emitter leakage current flows:

$$I_{CEO} = (\beta + 1)I_{CBO}$$

Common Collector (CC) Configuration (Emitter Follower)

- **Input Terminal:** Base
- **Output Terminal:** Emitter
- **Common Terminal:** Collector

1. Input Characteristics (I_B vs. V_{BC} at constant V_{EC})

- The input voltage V_{BC} depends heavily on the output voltage V_{EC} because $V_{BC} = V_{BE} - V_{CE}$.

2. Output Characteristics (I_E vs. V_{EC} at constant I_B)

- Similar in profile to the CE output curves, but the output current is $I_E = (\beta + 1)I_B \approx I_C$.
- Characterized by high input impedance, low output impedance, and a voltage gain slightly less than unity ($A_v \approx 1$). It is widely utilized as a buffer stage and impedance-matching amplifier.

Performance Comparison of BJT Configurations

Parameter	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Input Impedance (R_{in})	Low ($30\ \Omega - 100\ \Omega$)	Moderate ($1\ \text{k}\Omega - 5\ \text{k}\Omega$)	Very High ($20\ \text{k}\Omega - 500\ \text{k}\Omega$)
Output Impedance (R_{out})	Very High ($500\ \text{k}\Omega - 2\ \text{M}\Omega$)	Moderate ($10\ \text{k}\Omega - 50\ \text{k}\Omega$)	Very Low ($20\ \Omega - 100\ \Omega$)
Voltage Gain (A_v)	High ($100 - 500$)	High ($100 - 500$)	Less than 1 (≈ 0.99)
Current Gain (A_i)	Less than 1 ($\alpha \approx 0.98$)	High ($\beta \approx 50 - 300$)	High ($\beta + 1$)
Power Gain (A_p)	Moderate	Very High	Low to Moderate
Phase Shift (V_{out} vs V_{in})	0° (In-phase)	180° (Out-of-phase)	0° (In-phase)
Primary Use Case	High-frequency RF	General Voltage Amp	Buffer / Impedance Matching

3. Transistor DC Biasing and Stability

Biasing establishes a stable, predetermined DC operating point (**Q-point:** I_{CQ} , V_{CEQ}) in the active region of the transistor's characteristic curves so that AC input signals can be amplified symmetrically without entering

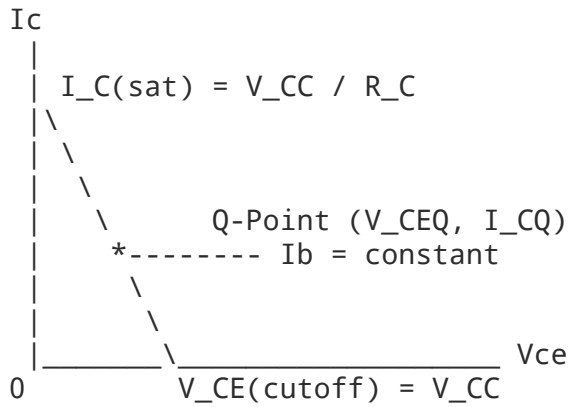
saturation or cutoff.

DC Load Line and Q-Point

Consider a standard CE output loop:

$$V_{CC} = I_C R_C + V_{CE} \implies I_C = -\frac{1}{R_C} V_{CE} + \frac{V_{CC}}{R_C}$$

- **Y-Intercept (Saturation Point):** Set $V_{CE} = 0 \implies I_{C(sat)} = \frac{V_{CC}}{R_C}$
- **X-Intercept (Cutoff Point):** Set $I_C = 0 \implies V_{CE(cutoff)} = V_{CC}$
- **Operating Point (Q):** The intersection of the DC load line with the specific base current curve ($I_B = I_{BQ}$).



Stability Factor (S)

The operating point shifts due to variations in:

1. Temperature (T), which causes I_{CBO} to double every 10°C rise.
2. Base-to-emitter voltage V_{BE} (decreases by approximately $2.5 \text{ mV}/^\circ\text{C}$).
3. Current gain β (varies widely with transistor manufacturing tolerances and temperature).

The primary stability factor is defined as:

$$S = \left. \frac{\partial I_C}{\partial I_{CBO}} \right|_{V_{BE}, \beta = \text{const}}$$

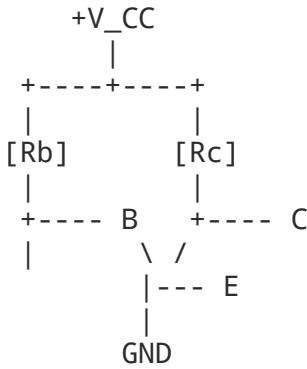
Using the relation $I_C = \beta I_B + (\beta + 1) I_{CBO}$, differentiating with respect to I_C yields:

$$S = \frac{1 + \beta}{1 - \beta \left(\frac{\partial I_B}{\partial I_C} \right)}$$

An ideal bias network achieves $S \rightarrow 1$, indicating that I_C is completely independent of variations in I_{CBO} .

Fixed Bias (Base Bias) Circuit

In this circuit, a single resistor R_B is connected directly between the DC supply V_{CC} and the base terminal.



Circuit Analysis

1. Input Loop (Base-Emitter):

$$V_{CC} - I_B R_B - V_{BE} = 0 \implies I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

2. Collector Current:

$$I_C = \beta I_B = \beta \left(\frac{V_{CC} - V_{BE}}{R_B} \right)$$

3. Output Loop (Collector-Emitter):

$$V_{CC} - I_C R_C - V_{CE} = 0 \implies V_{CE} = V_{CC} - I_C R_C$$

Stability Factor

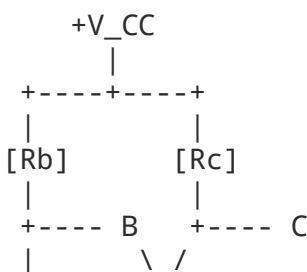
Since $I_B = \frac{V_{CC} - V_{BE}}{R_B}$, $\frac{\partial I_B}{\partial I_C} = 0$. Substituting into the general equation:

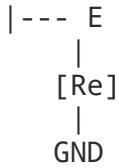
$$S = 1 + \beta$$

- **Verdict:** Highly unstable. Since β is large (e.g., 100), $S \approx 101$. Any thermal drift in I_{CBO} is amplified by $\beta + 1$, leading directly to thermal runaway and shifting of the Q-point. It is rarely used where stability is required.

Emitter Bias (Self-Bias / Emitter Resistor Circuit)

To stabilize the circuit, an emitter resistor R_E is added. It introduces negative feedback: an increase in I_C increases I_E , raising the voltage $V_E = I_E R_E$, which in turn reduces V_{BE} and lowers I_B , counteracting the original increase.





Circuit Analysis

1. Input Loop:

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

Using $I_E = (\beta + 1)I_B$:

$$V_{CC} - V_{BE} - I_B [R_B + (\beta + 1)R_E] = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

2. Collector Current:

$$I_C = \beta I_B = \frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E}$$

3. Output Loop:

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \quad (\text{assuming } I_E \approx I_C)$$

Stability Factor

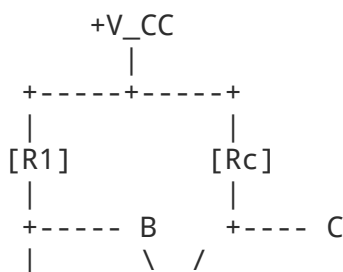
From the input loop: $I_B = \frac{V_{CC} - V_{BE} - I_C R_E}{R_B + R_E} \implies \frac{\partial I_B}{\partial I_C} = -\frac{R_E}{R_B + R_E}$.

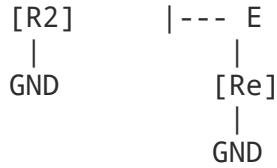
$$S = \frac{1 + \beta}{1 + \beta \left(\frac{R_E}{R_B + R_E} \right)}$$

- **Verdict:** If R_E is sufficiently large, S drops well below $1 + \beta$, providing substantially improved thermal stability.

Voltage Divider Bias (Universal Bias Circuit)

This is the most common biasing circuit. Resistors R_1 and R_2 act as a potential divider across V_{CC} , setting a fixed base voltage.





Exact Analysis Using Thévenin's Theorem

Replace the voltage divider (V_{CC} , R_1 , R_2) at the base terminal with its Thévenin equivalent:

$$V_{th} = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

1. Input Loop:

$$V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = (\beta + 1)I_B$:

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1)R_E}$$

2. Collector Current:

$$I_C = \frac{\beta(V_{th} - V_{BE})}{R_{th} + (\beta + 1)R_E}$$

3. Output Loop:

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Approximate Analysis (Valid if $\beta R_E \geq 10R_2$)

If the base current is negligible compared to the current through the divider:

$$V_B \approx V_{th} = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_E = V_B - V_{BE}$$

$$I_C \approx I_E = \frac{V_E}{R_E} = \frac{V_B - V_{BE}}{R_E}$$

- **Key Advantage:** I_C is completely independent of β .

Stability Factor

$$S = (1 + \beta) \cdot \frac{1 + \frac{R_{th}}{R_E}}{1 + \beta + \frac{R_{th}}{R_E}}$$

If $\frac{R_{th}}{R_E} \ll 1$, then $S \rightarrow 1$, delivering near-ideal stability against temperature and transistor variations.

4. Field Effect Transistors (FET)

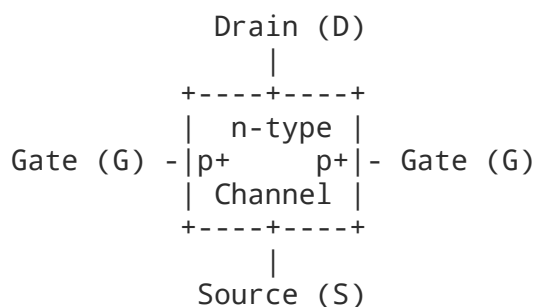
A Field Effect Transistor is a **unipolar** device; conduction relies exclusively on either electrons (in n-channel) or holes (in p-channel). It is **voltage-controlled**: the voltage applied to the gate modulates an electric field that controls the channel conductivity and drain current.

Comparison: BJT vs. FET

Characteristic	BJT	FET (JFET / MOSFET)
Carrier Type	Bipolar (electrons + holes)	Unipolar (electrons or holes)
Control Mechanism	Current-controlled ($I_C = f(I_B)$)	Voltage-controlled ($I_D = f(V_{GS})$)
Input Impedance	Low to Moderate (1 k Ω – 10 k Ω)	Extremely High (10 ⁸ Ω for JFET, 10 ¹⁴ Ω for MOSFET)
Noise Figure	Higher noise	Lower noise
Thermal Stability	Negative temp coefficient (susceptible to thermal runaway)	Positive temp coefficient (avoids thermal runaway)
Fabrication Size	Larger physical footprint on silicon	Extremely small (ideal for VLSI integration)

Junction Field Effect Transistor (JFET)

An n-channel JFET consists of an *n*-type silicon bar with two *p*⁺-type regions diffused on opposite sides to form the **gate**.



Operation

1. $V_{GS} = 0$ V, **Increasing** $V_{DS} > 0$:

- A voltage across Drain-to-Source drives electrons from Source to Drain.
- Because the drain is at a higher positive potential than the source, the *pn* junctions become increasingly reverse-biased near the drain end.
- The depletion regions widen progressively toward the drain end, narrowing the channel.

2. Pinch-Off ($V_{DS} = V_P$):

- At $V_{DS} = |V_P|$, the depletion layers approach each other, leaving a narrow saturation channel.
- Further increases in V_{DS} cannot pull more current; the drain current saturates at its maximum value: I_{DSS} (Drain-to-Source Saturation Current with gate shorted to source).

3. $V_{GS} < 0$:

- Applying a negative gate bias adds directly to the reverse bias across the junction.
- The channel pinches off at lower values of V_{DS} .
- When $V_{GS} = V_{GS(off)} = -V_P$, the channel is completely depleted, and $I_D = 0$.

JFET Characteristics

1. Drain Characteristics (I_D vs. V_{DS} at constant V_{GS})

Consists of three operational zones:

1. **Ohmic / Linear Region** ($V_{DS} < V_{DS(sat)} = V_{GS} - V_P$): Channel acts as a voltage-variable resistor (r_{ds}).
2. **Saturation / Active / Pinch-off Region** ($V_{DS} \geq V_{GS} - V_P$): I_D is essentially constant for a fixed V_{GS} .
3. **Breakdown Region**: Very high V_{DS} induces avalanche breakdown of the gate-channel junction, resulting in an uncontrolled surge in current.

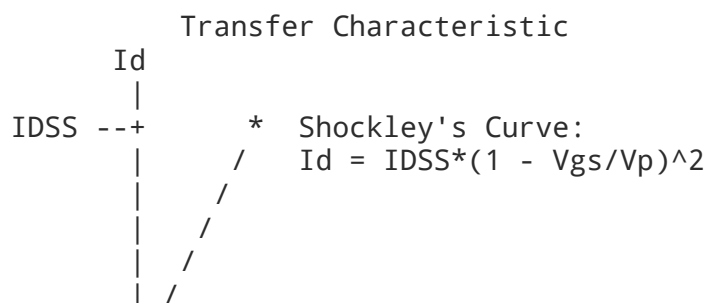
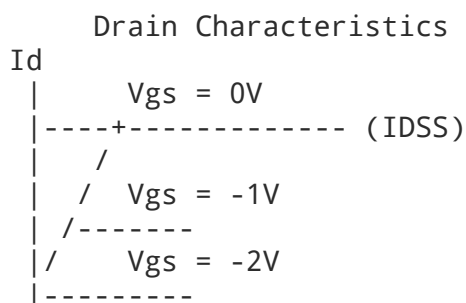
2. Transfer Characteristics (I_D vs. V_{GS})

The relationship between output current I_D and input voltage V_{GS} in the saturation region follows **Shockley's Equation**:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Where:

- I_{DSS} is the maximum saturation drain current at $V_{GS} = 0$ V.
- V_P is the pinch-off voltage ($V_P < 0$ for n-channel; $V_P > 0$ for p-channel).
- V_{GS} is the applied gate-to-source voltage.



+----- V_{ds}
0 V_p (Pinch-off)

+-----+----- V_{gs}
V_p 0

Transconductance (g_m)

Transconductance measures control sensitivity:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = g_{m0} \left(1 - \frac{V_{GS}}{V_P}\right)$$

Where $g_{m0} = \frac{2I_{DSS}}{|V_P|}$ is the maximum transconductance (at $V_{GS} = 0$ V).

Metal-Oxide-Semiconductor FET (MOSFET)

MOSFETs eliminate gate conduction by isolating the metallic (or polysilicon) gate terminal from the channel using an ultra-thin dielectric layer of Silicon Dioxide (SiO₂). As a result, input DC gate currents are virtually zero ($R_{in} \approx 10^{12} - 10^{15} \Omega$).

1. Depletion-Type MOSFET (D-MOSFET)

- **Construction:** Contains a physically diffused physical channel between the drain and source during manufacturing.
- **Modes of Operation:**
 - **Depletion Mode:** Apply negative V_{GS} (for n-channel). The negative gate repels conduction electrons out of the channel, depleting it and reducing I_D .
 - **Enhancement Mode:** Apply positive V_{GS} (for n-channel). The positive gate attracts additional electrons from the substrate into the channel, enhancing conductivity and raising $I_D > I_{DSS}$.
- **Equation:** Follows Shockley's equation for both positive and negative V_{GS} :

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

2. Enhancement-Type MOSFET (E-MOSFET)

- **Construction:** Has **no physical channel** between Drain and Source at fabrication. The substrate extends directly to the SiO₂ insulating layer.
- **Threshold Voltage (V_{th} or V_T):** The minimum positive gate-to-source voltage required to invert the substrate surface and create a conducting channel.
- **Operation (n-channel):**
 1. $V_{GS} < V_T$: No channel exists; back-to-back pn junctions prevent current flow ($I_D \approx 0$).
 2. $V_{GS} \geq V_T$: Positive gate charges repel holes in the p -substrate downward and attract minority electrons upward to the surface immediately under the SiO₂ layer. This forms an **inversion layer** (n -type channel) interconnecting Source and Drain.

- **Current Equations in E-MOSFET:**

- **Non-Saturation (Ohmic/Triode) Region** ($V_{DS} < V_{GS} - V_T$):

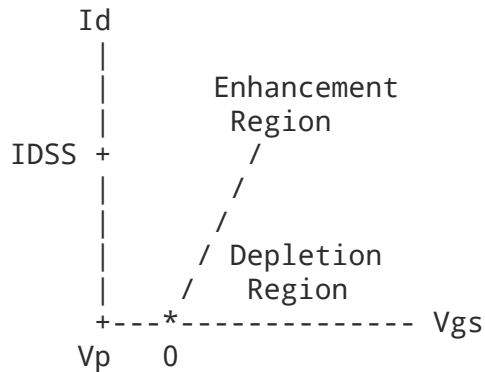
$$I_D = k_n [2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

- **Saturation Region** ($V_{DS} \geq V_{GS} - V_T$): The channel pinches off at the drain end:

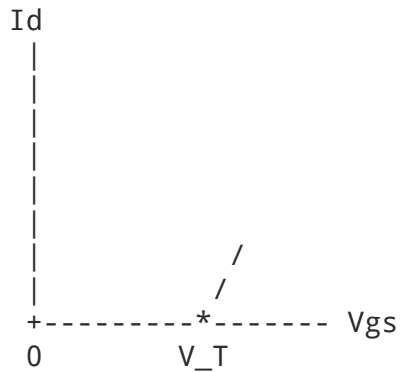
$$I_D = k_n(V_{GS} - V_T)^2$$

Where $k_n = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)$ is the conduction parameter (μ_n : mobility, C_{ox} : oxide capacitance, W/L : transistor channel width-to-length ratio).

D-MOSFET Transfer Curve

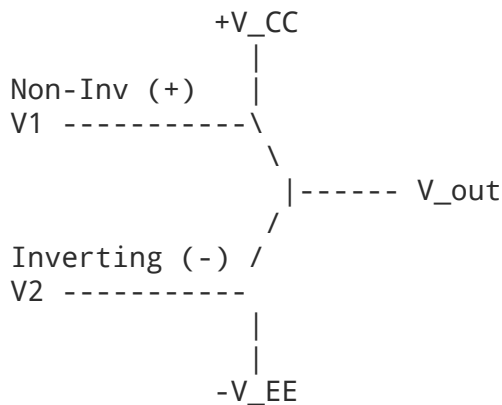


E-MOSFET Transfer Curve



5. Operational Amplifiers (Op-Amps) and Applications

An Operational Amplifier is a direct-coupled, high-gain integrated differential voltage amplifier with two input terminals and a single ground-referenced output.



Ideal vs. Practical Op-Amp Parameters (e.g., LM741)

Parameter	Symbol	Ideal Value	Practical Value (LM741)
Open Loop Voltage Gain	A_{OL}	∞	$\approx 2 \times 10^5$ (106 dB)
Input Impedance	R_{in}	∞	$\approx 2 \text{ M}\Omega$ (Bipolar) / $10^{12} \Omega$ (FET)
Output Impedance	R_{out}	0Ω	$\approx 75 \Omega$

Parameter	Symbol	Ideal Value	Practical Value (LM741)
Bandwidth	BW	∞	$\approx 1 \text{ MHz}$ (Unity Gain BW)
Input Offset Voltage	V_{os}	0 V	1 mV – 5 mV
Common Mode Rejection Ratio	$CMRR$	∞	$\approx 90 \text{ dB}$
Slew Rate	SR	∞	$\approx 0.5 \text{ V}/\mu\text{s}$

Core Principles for Linear Op-Amp Analysis

When negative feedback is present:

1. **Virtual Short Principle:** Because $A_{OL} \rightarrow \infty$ and $V_{out} = A_{OL}(V_+ - V_-)$, a finite output voltage requires:

$$V_+ - V_- = \frac{V_{out}}{A_{OL}} \rightarrow 0 \implies V_+ \approx V_-$$

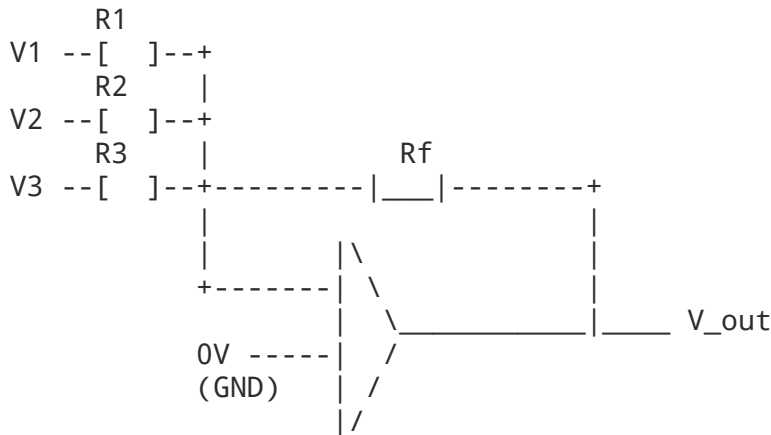
2. **Virtual Ground:** If the non-inverting terminal (V_+) is physically connected to ground (0 V), the inverting terminal (V_-) sits at approximately 0 V, acting as a "virtual ground."

3. **Zero Input Terminal Current:** Because $R_{in} \rightarrow \infty$, no current enters either input pin:

$$I_+ = I_- = 0$$

1. Inverting Adder (Summing Amplifier)

The summing amplifier accepts multiple input voltages and produces an output proportional to their negative weighted sum.



Derivation

Apply Kirchhoff's Current Law (KCL) at node V_- :

$$\sum I_{in} = I_f$$

$$I_1 + I_2 + I_3 = I_f$$

Expressing branch currents in terms of node voltages:

$$\frac{V_1 - V_-}{R_1} + \frac{V_2 - V_-}{R_2} + \frac{V_3 - V_-}{R_3} = \frac{V_- - V_{out}}{R_f}$$

By the virtual ground concept, $V_+ = 0 \text{ V} \implies V_- = 0 \text{ V}$:

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_f}$$

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

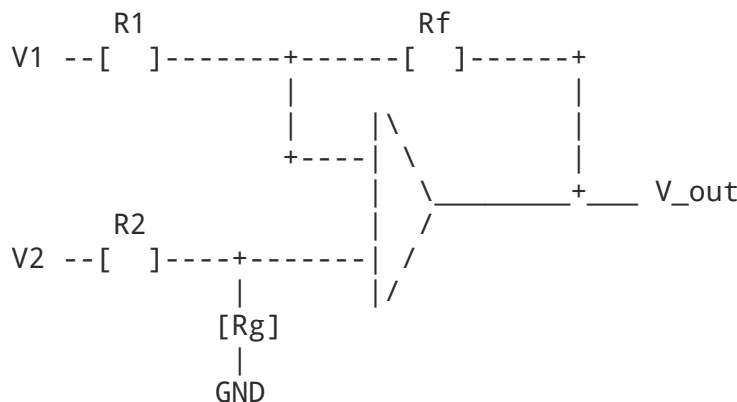
If $R_1 = R_2 = R_3 = R$:

$$V_{out} = -\frac{R_f}{R}(V_1 + V_2 + V_3)$$

If additionally $R_f = R$, it functions as an exact inverting summer: $V_{out} = -(V_1 + V_2 + V_3)$.

2. Subtractor (Difference Amplifier)

A subtractor produces an output proportional to the algebraic difference between two input signals: $V_2 - V_1$.



Derivation

1. **Find voltage at the non-inverting terminal (V_+):** Using the voltage divider rule (since $I_+ = 0$):

$$V_+ = V_2 \left(\frac{R_g}{R_2 + R_g} \right)$$

2. **Apply KCL at the inverting node (V_-):**

$$\frac{V_1 - V_-}{R_1} = \frac{V_- - V_{out}}{R_f}$$

$$V_{out} = V_- \left(1 + \frac{R_f}{R_1} \right) - V_1 \left(\frac{R_f}{R_1} \right)$$

3. **Use the virtual short condition ($V_- = V_+$):**

$$V_{out} = \left[V_2 \left(\frac{R_g}{R_2 + R_g} \right) \right] \left(1 + \frac{R_f}{R_1} \right) - V_1 \left(\frac{R_f}{R_1} \right)$$

$$V_{out} = V_2 \left(\frac{R_g}{R_2 + R_g} \right) \left(\frac{R_1 + R_f}{R_1} \right) - V_1 \left(\frac{R_f}{R_1} \right)$$

4. **Balanced Bridge Condition:** Set the resistor ratios such that $\frac{R_f}{R_1} = \frac{R_g}{R_2}$:

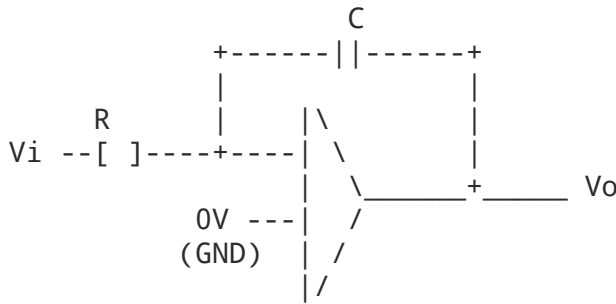
$$V_{out} = \frac{R_f}{R_1} (V_2 - V_1)$$

If $R_1 = R_2 = R_f = R_g$:

$$V_{out} = V_2 - V_1$$

3. Op-Amp Integrator

An integrator produces an output voltage proportional to the time integral of the input signal.



Derivation

Apply KCL at the inverting node (V_-):

$$I_R = I_C$$

From resistor and capacitor branch equations:

$$I_R = \frac{V_i - V_-}{R}$$

$$I_C = C \frac{d(V_- - V_o)}{dt}$$

Since $V_+ = 0$ V, the virtual ground gives $V_- = 0$ V:

$$\frac{V_i}{R} = C \frac{d(-V_o)}{dt} = -C \frac{dV_o}{dt}$$

$$\frac{dV_o}{dt} = -\frac{1}{RC} V_i(t)$$

Integrating both sides with respect to time from $t = 0$ to t :

$$V_o(t) = -\frac{1}{RC} \int_0^t V_i(\tau) d\tau + V_o(0)$$

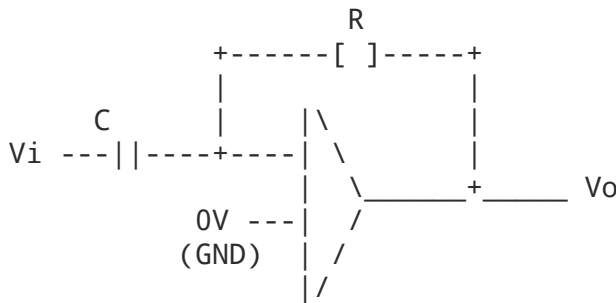
Where $V_o(0)$ is the initial voltage on the feedback capacitor.

Practical Considerations

- **Frequency Domain:** The gain is $|A_v| = \frac{1}{\omega RC}$. At DC ($\omega \rightarrow 0$), the gain approaches infinity ($A_v \rightarrow \infty$). Any minor DC offset voltage or input bias current will saturate the op-amp output rail-to-rail over time.
- **Practical (Lossy) Integrator:** A large feedback resistor R_f is placed in parallel with C . This caps the DC closed-loop gain to $-R_f/R$, stabilizing the circuit against drift.

4. Op-Amp Differentiator

A differentiator produces an output voltage proportional to the instantaneous time rate of change of the input signal.



Derivation

Apply KCL at the inverting node (V_-):

$$I_C = I_R$$

Using element equations:

$$I_C = C \frac{d(V_i - V_-)}{dt}$$

$$I_R = \frac{V_- - V_o}{R}$$

Since $V_+ = 0$ V, virtual ground enforces $V_- = 0$ V:

$$C \frac{dV_i}{dt} = \frac{-V_o}{R}$$

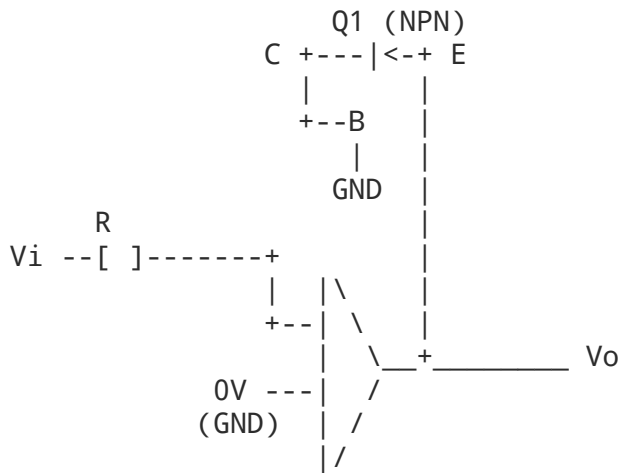
$$V_o(t) = -RC \frac{dV_i(t)}{dt}$$

Practical Considerations

- **Instability and Noise:** The differentiator gain is $|A_v| = \omega RC$. High-frequency noise components (such as RF pickup or ripple) are heavily magnified because gain increases proportionally with frequency (ω), which can drive the op-amp into instability or oscillation.
- **Practical Differentiator:** A small series resistor R_1 is added at the input, and a small bypass capacitor C_f is placed in parallel with R . This creates a band-limited differentiator that rolls off gain at high frequencies.

5. Logarithmic Amplifier (Log Amp)

A Logarithmic Amplifier produces an output voltage proportional to the natural logarithm of the input voltage. It utilizes the exponential current-voltage characteristic of a bipolar junction transistor (or diode) placed in the feedback loop.



Derivation

1. **Input Current:** Given $V_- = 0$ V (virtual ground), the input current through resistor R is:

$$I_{in} = \frac{V_i}{R}$$

2. **BJT Feedback Current:** The base of NPN transistor Q_1 is grounded ($V_B = 0$ V), and its emitter is connected to the output ($V_E = V_o$). Hence:

$$V_{BE} = V_B - V_E = 0 - V_o = -V_o$$

From the Ebers-Moll equation for a forward-biased base-emitter junction:

$$I_C = I_S \left(e^{\frac{V_{BE}}{\eta V_T}} - 1 \right) \approx I_S e^{\frac{V_{BE}}{\eta V_T}}$$

Where:

- I_S is the reverse saturation current of the transistor junction.
- $V_T = \frac{kT}{q}$ is the thermal voltage (≈ 26 mV at 300 K).
- η is the ideality factor ($\eta = 1$ for modern silicon planar transistors).

3. **Equate Currents via KCL at node V_- :**

$$I_{in} = I_C$$

$$\frac{V_i}{R} = I_S e^{\frac{-V_o}{V_T}}$$

4. **Solve for V_o :** Take the natural logarithm of both sides:

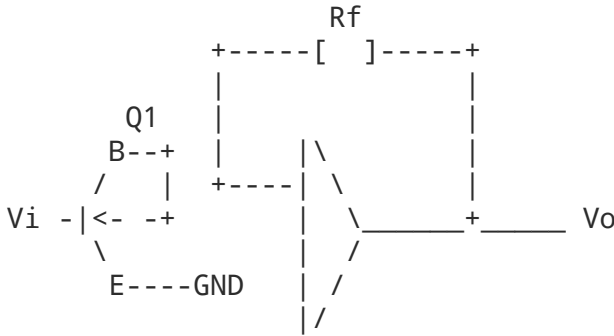
$$\ln \left(\frac{V_i}{RI_S} \right) = -\frac{V_o}{V_T}$$

$$V_o = -V_T \ln \left(\frac{V_i}{RI_S} \right)$$

The output voltage is directly proportional to the logarithm of the input voltage V_i .

6. Antilogarithmic Amplifier (Exponential Amp)

An Antilog (or Exponential) amplifier performs the inverse operation: its output voltage is proportional to the exponential of the input voltage (e^{kV_i}). The positions of the input resistor and feedback transistor are swapped.



Derivation

1. **Input Current through Transistor Q_1 :** The emitter is tied to ground ($V_E = 0$ V), and the base is driven by the input voltage:

$$V_B = V_i \implies V_{BE} = V_i - 0 = V_i$$

The collector current flowing into the inverting summing node is:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}} = I_S e^{\frac{V_i}{V_T}}$$

2. **Current Through Feedback Resistor R_f :** Since $V_- = 0$, all of I_C must pass through R_f :

$$I_f = I_C$$

3. **Output Voltage Expression:** With $V_- = 0$ V (virtual ground):

$$I_f = \frac{V_- - V_o}{R_f} = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}$$

Equating currents:

$$-\frac{V_o}{R_f} = I_S e^{\frac{V_i}{V_T}}$$

$$V_o = -R_f I_S e^{\frac{V_i}{V_T}}$$

Using the relation $e^x = 10^{\frac{x}{\ln(10)}} \approx 10^{\frac{x}{2.303}}$:

$$V_o = -R_f I_S \cdot 10^{\left(\frac{V_i}{2.303 V_T}\right)}$$

The circuit provides an output voltage directly proportional to the antilog (base e or base 10) of the input signal.

Operational Summary of Op-Amp Circuits

Circuit Application	Configuration Transfer Equation	Key Design Criterion
Summing Amp	$V_o = -R_f \sum \frac{V_k}{R_k}$	Summing node acts as virtual ground ($I_- = 0$).
Difference Amp	$V_o = \frac{R_f}{R_1} (V_2 - V_1)$	Balanced bridge condition: $\frac{R_f}{R_1} = \frac{R_g}{R_2}$.
Integrator	$V_o = -\frac{1}{RC} \int V_i dt$	Requires feedback resistor R_f in parallel with C for DC stability.
Differentiator	$V_o = -RC \frac{dV_i}{dt}$	Requires series input resistor R_{in} to attenuate high-frequency noise.
Log Amplifier	$V_o = -V_T \ln \left(\frac{V_i}{R I_S} \right)$	BJT in feedback path; requires temperature compensation for V_T and I_S .
Antilog Amplifier	$V_o = -R_f I_S e^{\frac{V_i}{V_T}}$	BJT at input path; limited to unipolar inputs to keep junction forward-biased.